

Setup:

Some rational functions are hard to integrate, but others are simple. Take

$$\int \frac{2}{x-1} dx = 2 \ln|x-1| + C \quad \text{and}$$

$$\int \frac{1}{x+2} dx = \ln|x+2| + C$$

but $\int \frac{x+5}{x^2+x-2} dx$ is difficult.

However, we notice

$$\int \frac{x+5}{x^2+x-2} dx = \int \frac{2}{x-1} dx - \int \frac{1}{x+2} dx$$

$$= 2 \ln|x-1| - \ln|x+2| + C$$

In this section, we learn how to simplify rational functions to a sum of simpler rational functions. This is called partial fraction decomposition.

Polynomial Division

Take $f(x) = \frac{P(x)}{Q(x)}$, P, Q Polynomials

highest power of x in $P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$

If $\deg(P) \geq \deg(Q)$ we need to simplify before conducting Partial fraction decomposition.

This will require Polynomial Division and we get

$$f(x) = S(x) + \frac{R(x)}{Q(x)} \quad \text{where } S(x) \text{ is a polynomial}$$

remainder

which is easy to integrate

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Note: Do polynomial division every time $\deg(P(x)) \geq \deg(Q(x))$ and only this time.

Ex:

$$\int \frac{x^3 + x}{x-1} dx, \text{ Notice } \deg(x^3 + x) = 3 \geq 1 = \deg(x-1)$$

So we do polynomial division.

To do Polynomial division, we will iteratively subtract multiples of the denominator from the numerator, keeping track of what we multiplied by.

$$\begin{array}{r}
 \boxed{x^2 + x + 2} \\
 x-1 \overline{) x^3 + 0x^2 + x + 0} \\
 \underline{-(x^3 - x^2)} \quad \text{multiply by } x^2 \text{ to match first term} \\
 x^2 + x + 0 \\
 \underline{-(x^2 - x)} \\
 2x + 0 \\
 \underline{-(2x - 2)} \\
 2 \quad \text{remainder}
 \end{array}$$

fill in 0's to help

$$\int \frac{x^3 + x}{x-1} dx = \int x^2 + x + 2 + \frac{2}{x-1} dx$$

$$= \frac{x^3}{3} + \frac{x^2}{2} + 2x + 2 \ln|x-1| + C$$

Now, cases where the denominator is simple are quick, but even with complicated denominators we can still simplify.

Some cases:

$$1) f(x) = \frac{P(x)}{(a_1x+b_1)(a_2x+b_2)\dots(a_nx+b_n)} = \frac{\overset{\text{Some number}}{A_1}}{a_1x+b_1} + \frac{A_2}{a_2x+b_2} + \dots + \frac{A_n}{a_nx+b_n}$$

2)

$$f(x) = \frac{p(x)}{(ax+b)^n} = \frac{A_1}{ax+b} + \frac{A_2}{(ax+b)^2} + \dots + \frac{A_n}{(ax+b)^n}$$

3)

$$f(x) = \frac{p(x)}{\underbrace{ax^2+bx+c}_{\text{irreducible, factor if you can}}} = \frac{Ax+b}{ax^2+bx+c} \text{ — one less power than denominator}$$

likewise

$$\frac{p(x)}{ax^3+bx^2+cx+d} = \frac{Ax^2+Bx+C}{ax^3+bx^2+cx+d}$$

Note: Combinations of these apply.

So

$$\frac{p(x)}{(ax+b)^2(cx^2+dx+e)} = \frac{A}{ax+b} + \frac{B}{(ax+b)^2} + \frac{Cx+D}{cx^2+dx+e}$$

Now, how do we find the constants in the numerator?

Answer: Lot of algebra.

Ex:

$$\int \frac{x^2+2x-1}{2x^3+3x^2-2x} dx$$

$$\frac{x^2+2x-1}{2x^3+3x^2-2x} = \frac{x^2+2x-1}{x(2x-1)(x+2)} = \frac{A}{x} + \frac{B}{2x-1} + \frac{C}{x+2}$$

Multiply by denominator on both sides

$$x^2 + 2x - 1 = \left(\frac{A}{x} + \frac{B}{2x-1} + \frac{C}{x+2} \right) x(2x-1)(x+2)$$

$$= A(2x-1)(x+2) + Bx(x+2) + Cx(2x-1)$$

distribute

$$= A(2x^2 + 3x - 2) + B(x^2 + 2x) + C(2x^2 - x)$$

$$= 2Ax^2 + 3Ax - 2A + Bx^2 + 2Bx + 2Cx^2 - Cx$$

group like terms

$$= 2Ax^2 + Bx^2 + 2Cx^2 + 3Ax + 2Bx - Cx - 2A$$

$$1x^2 + 2x - 1 = (2A + B + 2C)x^2 + (3A + 2B - C)x - 2A$$

match like coefficients

$$\begin{cases} 2A + B + 2C = 1 \\ 3A + 2B - C = 2 \\ -2A = -1 \rightarrow A = \frac{1}{2} \end{cases}$$

$$\begin{cases} 1 + B + 2C = 1 \rightarrow B = -2C \\ \frac{3}{2} + 2B - C = 2 \rightarrow \frac{3}{2} - 4C - C = 2 \\ -5C = \frac{1}{2} \end{cases}$$

$$C = -\frac{1}{10}$$

$$2C = B = -\frac{1}{5}$$

$$\int \frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x} dx = \int \frac{1}{2} \cdot \frac{1}{x} - \frac{1}{5} \frac{1}{x-1} - \frac{1}{10} \frac{1}{x+2} dx$$

$$= \frac{1}{2} \ln|x| - \frac{1}{10} \ln|2x-1| - \frac{1}{10} \ln|x+2| + C$$

$$\underline{\text{Ex:}} \int \frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} dx$$

Polynomial division first

$$\begin{array}{r} x^3 - x^2 - x + 1 \overline{) x^4 + 0x^3 - 2x^2 + 4x + 1} \\ \underline{-(x^4 - x^3 - x^2 + x)} \\ x^3 - x^2 + 3x + 1 \\ \underline{-(x^3 - x^2 - x + 1)} \\ 4x \end{array}$$

$$\frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} = x + 1 + \underbrace{\frac{4x}{x^3 - x^2 - x + 1}}_{\text{Partial fraction}}$$

$$\frac{4x}{x^3 - x^2 - x + 1} = \frac{4x}{(x-1)^2(x+1)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+1}$$

$$4x = A(x-1)(x+1) + B(x+1) + C(x-1)^2$$

$$4x = Ax^2 - A + Bx + B + Cx^2 - 2Cx + C$$

$$4x = (A+C)x^2 + (B-2C)x + (-A+B+C)$$

$$\begin{cases} A+C=0 & \leadsto A=-C \\ B-2C=4 \\ -A+B+C=0 \end{cases} \quad \downarrow$$

$$\leadsto B+2C=0 \rightarrow 2C=-B$$

$$B - 2C = 4 \rightarrow B + B = 4 \rightarrow \boxed{B = 2}$$

$$-2C = B \rightarrow \boxed{C = -1}$$

$$A = -C \rightarrow \boxed{A = 1}$$

$$\int \frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} = \int x + 1 + \frac{1}{x-1} + \frac{2}{(x-1)^2} - \frac{1}{x+1} dx$$

$$= \frac{1}{2}x^2 + x + \ln|x-1| - \frac{2}{x-1} - \ln|x+1| + C$$

$$\boxed{= \frac{1}{2}x^2 + x - \frac{2}{x-1} + \ln\left|\frac{x-1}{x+1}\right| + C}$$

Ex:

$$\int \frac{2x^2 - x + 4}{x^3 + 4x} dx$$

$$\frac{2x^2 - x + 4}{x^3 + 4x} = \frac{2x^2 - x + 4}{x(x^2 + 4)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 4}$$

$$2x^2 - x + 4 = A(x^2 + 4) + (Bx + C)x$$

$$= Ax^2 + 4A + Bx^2 + Cx$$

$$= (A+B)x^2 + Cx + 4A$$

$$\begin{cases} A+B = 2 & A=1 \\ C = -1 & \rightarrow B=1 \\ 4A = 4 & C=-1 \end{cases}$$

$$\int \frac{2x^2 - x + 4}{x^3 + 4x} dx = \int \frac{1}{x} + \frac{x-1}{x^2+4} dx$$

$$= \int \frac{1}{x} dx + \int \frac{x}{x^2+4} dx - \int \frac{1}{x^2+4} dx$$

$$= \ln|x| + \frac{1}{2} \ln(x^2+4) - \frac{1}{2} \arctan\left(\frac{x}{2}\right) + C$$

Ex: $\int \frac{4x^2 - 3x + 2}{4x^2 - 4x + 3} dx$

Polynomial division

$$\begin{array}{r} 1 \\ 4x^2 - 4x + 3 \overline{) 4x^2 - 3x + 2} \\ \underline{-(4x^2 - 4x + 3)} \\ x - 1 \end{array}$$

$$\int \frac{4x^2 - 3x + 2}{4x^2 - 4x + 3} dx = \int 1 + \underbrace{\frac{x-1}{4x^2 - 4x + 3}}_{\text{Partial fraction}} dx$$

Note: $4x^2 - 3x + 2$ is irreducible as $\sqrt{b^2 - 4ac}$ is not real.

$$\int 1 dx + \int \frac{x-1}{4x^2 - 4x + 2} dx = x + \int \frac{x-1}{4x^2 - 4x + 1 - 1 + 2} dx$$

$$= x + \int \frac{x-1}{(2x-1)^2 + 2} dx \quad \begin{array}{l} u = 2x-1 \rightarrow x = \frac{u+1}{2} \\ du = \frac{1}{2} dx \end{array}$$

$$= x + \frac{1}{2} \int \frac{\frac{u+1}{2}-1}{u^2+2} du = x + \frac{1}{4} \int \frac{u-1}{u^2+2} du$$

$$= x + \frac{1}{4} \int \frac{u}{u^2+2} du - \frac{1}{4} \int \frac{1}{u^2+2} du$$

$$= x + \frac{1}{8} \ln(u^2+2) - \frac{1}{4\sqrt{2}} \arctan\left(\frac{u}{\sqrt{2}}\right) + C$$

$$= x + \frac{1}{8} \ln(4x^2-4x+3) - \frac{1}{4\sqrt{2}} \arctan\left(\frac{2x-1}{\sqrt{2}}\right) + C$$

Ex: Write out Partial fraction form of

$$\frac{x^3 + x^2 + 1}{x(x-1)(x^2+x+1)(x^2+1)^3}$$

$$= \frac{A}{x} + \frac{B}{x-1} + \frac{Cx+D}{x^2+x+1} + \frac{Ex+F}{x^2+1} + \frac{Gx+H}{(x^2+1)^2} + \frac{Ix+J}{(x^2+1)^3}$$

If we did find the values of $A, \dots, I$ we get

$$= -\frac{A}{x} + \frac{1}{8x-8} - \frac{x+1}{x^2+x+1} + \frac{15x-1}{8x^2+8} + \frac{3x+3}{4(x^2+1)^2} - \frac{x-1}{2(x^2+1)^3}$$

Ex: $\int \frac{-x^3 + 2x^2 - x + 1}{x(x^2+1)^2} dx$

$$\frac{-x^3 + 2x^2 - x + 1}{x(x^2+1)^2} = \frac{A}{x} + \frac{Bx + C}{x^2+1} + \frac{Dx + E}{(x^2+1)^2}$$

$$\begin{aligned} -x^3 + 2x^2 - x + 1 &= A(x^2+1)^2 + (Bx+C)x(x^2+1) + (Dx+E)x \\ &= Ax^4 + 2Ax^2 + A + Bx^4 + Cx^3 + Bx^2 + Cx + Dx^2 + Ex \\ &= (A+B)x^4 + Cx^3 + (2A+B+D)x^2 + (C+E)x + A \end{aligned}$$

$$\begin{cases} A+B=0 \rightarrow \boxed{B=-1} \\ \boxed{C=-1} \\ 2A+B+D=2 \rightarrow \boxed{D=1} \\ C+E=-1 \rightarrow \boxed{E=0} \\ \boxed{A=1} \end{cases}$$

$$\int \frac{-x^3 + 2x^2 - x + 1}{x(x^2+1)^2} dx = \int \frac{1}{x} - \frac{x+1}{x^2+1} + \frac{x}{(x^2+1)^2} dx$$

$$= \int \frac{1}{x} dx - \int \frac{x+1}{x^2+1} dx + \int \frac{x}{(x^2+1)^2} dx$$

$$= \ln|x| - \int \frac{x}{x^2+1} dx - \int \frac{1}{x^2+1} dx + \int \frac{x}{(x^2+1)^2} dx$$

$$= \ln|x| - \frac{1}{2} \ln(x^2+1) - \arctan(x^2+1) - \frac{1}{2x^2+2} + C$$